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Analysis of photodiode and barrier properties of CoPc/n-Ge heterojunction under various illumination wavelengths

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A B S T R A C T

The present investigation reports the analysis of barrier parameters of the CoPc/n-Ge heterojunction photodiode under varied illumination wavelengths from 500 nm to 1200 nm in steps of 25 nm. The efficiency of this photodiode was also summarized using responsivity, detectivity and quantum efficiency for different illumination wavelengths. The photocurrent seems to be maximum for 850 nm and 1175 nm and shows higher responsivity of 1.72 A/W and 2.38 A/W at 775 nm and 1125 nm wavelength respectively. The quantum efficiency of the photodiode was found to be 275 % at illumination wavelength of 775 nm and 263 % at 1125 nm. For the selective illumination wavelengths (775 nm, 850 nm, 1125 nm and 1175 nm), the electrical properties of the CoPc/n-Ge photodiode were studied. Noticeable variation in the barrier parameters of the photodiode is observed with illumination than compared to dark. Significantly, the series resistance was found to be decreased and shunt resistance is increased with illumination wavelength than compared to dark. Current conduction mechanisms were also investigated in both the forward and reverse bias of the photodiode. The forward current transport processes show that at moderately high bias the current conduction might be associated with TCLC. It is also evidenced from the reverse conduction mechanism that at low reverse voltages PFE is found to be dominant whereas SE is dominant at high reverse voltages. The reduced interface state densities at the CoPc/n-Ge interface were observed under illumination than compared to dark.

1. Introduction

Fabrication of efficient photodiodes using organic materials has attracted great attention due to its advantages such as broad range of materials selection, low-cost processing methodologies, privilege in modification of chemical properties as well as lightweight and flexibility [1]. These advantages allow organic materials can be used effectively in a wide variety of applications such as defence, medical and memory devices. Further, the development of low-cost organic photodetector devices urges the exploration of new and composite organic materials. In general, photodetector devices were fabricated using different structures like Schottky junctions,

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metal-organic semiconductor junctions and bulk heterojunction thin films (either organic, inorganic or hybrid junctions) which usually form a p-n junction [2–4]. Among these various structures, p-n junction-based photodiodes show efficient photoresponse due to their excellent charge separation in the presence of built-in electric field. However, most of the photodiodes commercially available in the market utilize Si-based technology which is highly cost-effective and comes in larger volumes [3]. On the other hand, bulk heterojunction organic thin films defined with donor-acceptor molecules can efficiently enhance the photoresponsivity. These heterojunctions also eliminate the shortcomings of low charge carrier mobility and offer low permittivity of organic semiconductors [4]. Further, the barrier between donor and acceptor molecules provides efficient charge carrier separation and the built-in field makes the charge carriers to reach to the contact electrode without much recombination.

The photodiodes have been fabricated using hybrid heterojunctions such as organic-on-inorganic heterostructures. These hybrid junctions provide the advantages of both organic and inorganic substances where inorganic material provides greater thermal stability and organic molecules are empowered with superior chemical, biological compatibility and molecular electronic stability. Many reports are available on the fabrication and photodiode characterization of organic/Si-based heterojunction interfaces [5–9]. Cavdar et al. [5] reported the study of photodiode properties of triphenylamine/Si heterojunction under various illumination intensities. Pentacene/Si heterojunctions were fabricated using DC magnetron sputtering and analysed its photodiode properties at different illumination intensities [6]. Erdogan et al. [7] fabricated electron-rich organic thin films on n-Si and studied their photoresponsive properties under illumination. Si/perylene-derivative-based hybrid heterojunction was fabricated by Bednorz et al. [8] and studied its photoresponsive properties at near-infrared wavelength range in the illumination wavelength of 1.55 μm for telecom applications. ZnO-doped PCBM organic layer on Si was fabricated and studied its suitability to photoresponsive applications using I-V properties under various illumination intensities [9]. Cobalt centered-complexes and cobalt, nickel based polyoxometalates (POMs) were synthesized by chemical reaction and utilsied as an photoactive interlayer in the fabrication of p-Si heterojunction photodiode [10,11]. Still, the researchers are exploring various types of materials to fabricate Si-based heterojunctions and intended to understand their photoresponsive properties for photodiode applications.

On the other hand, a few recent reports are reviewed here on the fabrication and photodiode characterization of metal-phthalocyanine/Si heterojunctions [12–15]. Garzon-Roman et al. [12] utilized erbium-phthalocyanine as an interlayer on porous Si and studied its photoresponsive properties. They observed the photoresponsivity of these heterostructures at larger reverse voltages such as −10 V. Hamzah and Hassan [13] fabricated a FePc thin layer on n-Si with different thickness and observed an enhanced photoresponsivity for the FePc thin films covered with an organic polymer blend. Zhong et al. [14] fabricated a photodetector using CuPc on p-Si and studied its photoresponsive properties at 655 nm illumination wavelength. Graphene oxide (GO) doped CoPc thin film was formed on p-Si to fabricate a heterojunction and studied its photodiode properties for different GO concentrations and at different illumination intensities [15]. They observed that different GO concentration alters the distribution of trap levels which influences the photovoltaic properties.

Recently, germanium (Ge) semiconductors have gained importance over silicon (Si) semiconductors due to their superiority in electron and hole mobility at low electric fields [16,17]. But, research on Ge-based device technology was hindered due to Fermi-level pinning phenomena and thermal stability [18,19]. These challenging issues can be resolved by fabricating Ge-based Schottky junctions with an interlayer possibly with a thin oxide layer or by an interlayer preferably by an organic thin film. Presently, very few reports are available which discuss the photodiode properties of the n-Ge heterojunctions [20–23]. Zhu et al. [20] has studied the NIR photodetection properties of n-Ge-based heterojunctions using graphene and graphene derivatives. They observed superior photodetection properties like responsivity and detectivity at an illumination wavelength of 1550 nm. A high-quality n-Ge photodetector using MoS_2 as an interlayer was demonstrated and found highly photoresponsive from 300 nm to 1700 nm wavelength [21]. Lee et al. [22] proposed 2D/3D heterostructures for high-speed broad-band photodetectors using WSe_2 as 2D material and n-Ge as 3D material. They observed a strong photoresponsivity and detectivity at the visible region (520 nm) and infrared region (1550 nm). Recently, polarization sensitive photodetector was fabricated using black phosphorus on Ge and observed superior detectivity at 532 nm and 1550 nm [23]. To the best of our knowledge, there are no reports available on the study of photodetection properties of metal-phthalocyanines (MPcs) modified n-Ge heterostructures. MPcs are found to be an effective optoelectronic materials due to their better photodetectivity to illumination, physical and chemical stability [24]. Among the various types of MPcs, the cobalt phthalocyanine (CoPc) exhibited p-type conductivity with superior carrier mobility [25]. CoPc is utilized in various specific applications as humidity sensors and temperature sensors due to its exceptional properties such as thermal stability and hydrophobicity [26]. Hence a detailed investigation of photodetection properties of CoPc/n-Ge heterostructure is very much essential to know its applicability in optoelectronic devices.

The present work emphasizes the electrical properties of CoPc modified n-Ge heterostructures under illumination at various wavelengths. The photodiode properties such as photocurrent, responsivity, detectivity and quantum efficiency were analysed for the CoPc/Ge heterojunction in the wavelength range from 500 nm to 1200 nm in steps of 25 nm. The barrier properties of the heterostructure like barrier height, ideality factor, series resistance (R_S), shunt resistance (R_{sh}), different conduction processes and photoresponsivity were analysed at various selective wavelengths (775, 850, 1125, 1175 nm). The variation of R_S and R_{sh} significantly influences the photoresponse of the heterostructure at selective wavelengths. The forward conduction properties reveal that conduction through traps is very much dominant at moderately high voltage under illumination. The interface state densities of the heterojunction are found to be smaller under illumination than compared to dark.

2. Experimental details

In the current analysis, an n-Ge wafer (100) (Sb-doped) was employed with the carrier concentration of 5×10^{15} cm^{-3} was purchased from Semiconductor Wafer Inc., Taiwan. Firstly, the n-Ge wafer was cleaned with trichloroethylene, acetone and methanol using ultrasonication and further cleaned in H_2O: HF (100:1) solution to remove the native oxide layer on the n-Ge surface. Then, the wafer was dried in an N_2 atmosphere and the same is transferred to a vacuum coating unit to deposit low resistance ohmic contact (Al) of thickness 50 nm on the rough side of the sample. The CoPc substance was purchased from Tokyo chemical industries co. Ltd. having CAS NO. 3317–67–7 with the dye purity of >93 %. The CoPc interlayer is deposited on the polished side of the Ge wafer by using the thermal evaporation method. The Schottky metal (Au) has been evaporated by using electron beam evaporation with a shadow mask of 0.5 mm diameter. Photoresponsivity measurements of as-deposited CoPc/Ge heterostructures employed in a dark room equipped with a 2400 Keithley source meter, monochromator (make: Bentham SSM150Xe) attached to a xenon lamp of 75 W power and optical power sensor (make: Newport 918D–UV–0D3R) [27].

3. Results and discussion

Fig. 1 depicts the current-voltage (I-V) properties of the Au/CoPc/n-Ge heterostructures at various illumination wavelengths. The photocurrent plotted against forward voltage is presented in Fig. 1(a) whereas Fig. 1(b) illustrates the photocurrent versus reverse voltage. From Fig. 1(a) and (b), it is evident that the photocurrent was found to increase with illumination wavelength both in forward and reverse bias voltages. Increase in photocurrent under illumination might be due to charge carrier generation and migration across the barrier in the presence of high electric fields. In general observation, the photoresponsivity in forward bias voltages under illumination seems to be very poor due to the recombination of charge carriers at low electric fields. But in this heterostructure, there is a significant photoresponse observed at forward voltages which also indicates the lower recombination rate of charge carriers at the barrier. Similar properties of photoresponsivity at both forward and reverse voltages were observed in the earlier reports [20,28].

Fig. 2 reveals the photocurrent and responsivity properties of the CoPc/n-Ge heterostructure for different illumination wavelengths ranging from 500 nm to 1200 nm. The photocurrent of the heterostructure was analysed at −2 V reverse bias as shown in Fig. 2(a). The photocurrent measured at a bias of −2 V seems to be maximum at 850 nm (with a normalized power of 5.5 μW/cm^2) and 1175 nm (with a normalized power of 4.4 μW/cm^2). From the optical absorption spectra of CoPc thin films reported in one of our earlier publications [27], the maximum photocurrent observed at 850 nm might be associated with CoPc semiconductor. Further, the photon energy noted for 850 nm wavelength (1.46 eV) is nearly matched with the bandgap of CoPc [29]. On the other hand, the maximum photocurrent observed at 1175 nm can be related to the n-Ge semiconductor which initiates the excitation of carriers from the valence band to conduction band. The energy band diagram of CoPc/Ge heterojunction under illumination is included in Fig. 2(c).

Photosensitizing properties of any photodiode can be analysed quantitatively using key parameters such as responsivity, detectivity and external quantum efficiency. These quantities might be evaluated with the following equations [30]

$$\text{Responsivity}(\mathrm{R}) = \frac{\mathrm{I_{ph}} - \mathrm{I_{dark}}}{\mathrm{P} * \mathrm{A}} \tag{1}$$

$$\text{Detectivity}(\mathrm{D}) = \frac{\mathrm{R}*(\mathrm{A})^{1/2}}{(2 * \mathrm{q}*\mathrm{I_{dark}})^{1/2}} \tag{2}$$

$$\text{External quantum efficiency (EQE)} = \frac{\mathrm{R} * \mathrm{h} * \mathrm{c}}{\mathrm{q} * \lambda_{\mathrm{Incident}}} \tag{3}$$

where the terms included in the above expression like I_{ph}, I_{dark}, P, A, $\lambda_{Incident}$ represent the photocurrent and dark current at a particular reverse bias voltage, incident normalized power, active area of the junction and incident wavelength, respectively. The responsivity, detectivity and quantum efficiency of the CoPc/n-Ge heterostructure were presented in Figs. 2(b), 3(a) and (b) respectively for illumination wavelengths range from 500 nm to 1200 nm in steps of 25 nm. It is revealed from Fig. 2(b) that the

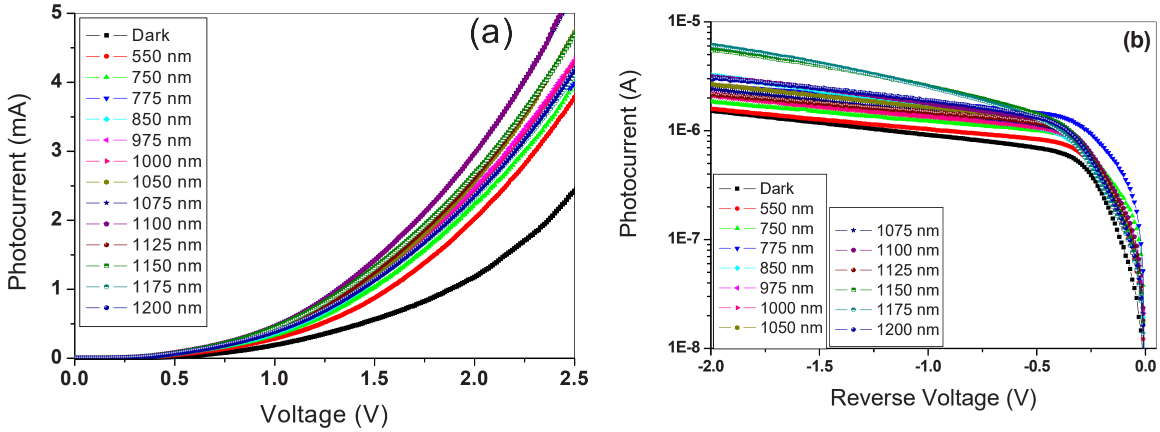

Fig. 1. (a) Photocurrent versus forward voltage and (b) photocurrent versus reverse voltage of CoPc/n-Ge heterojunction photodiode.

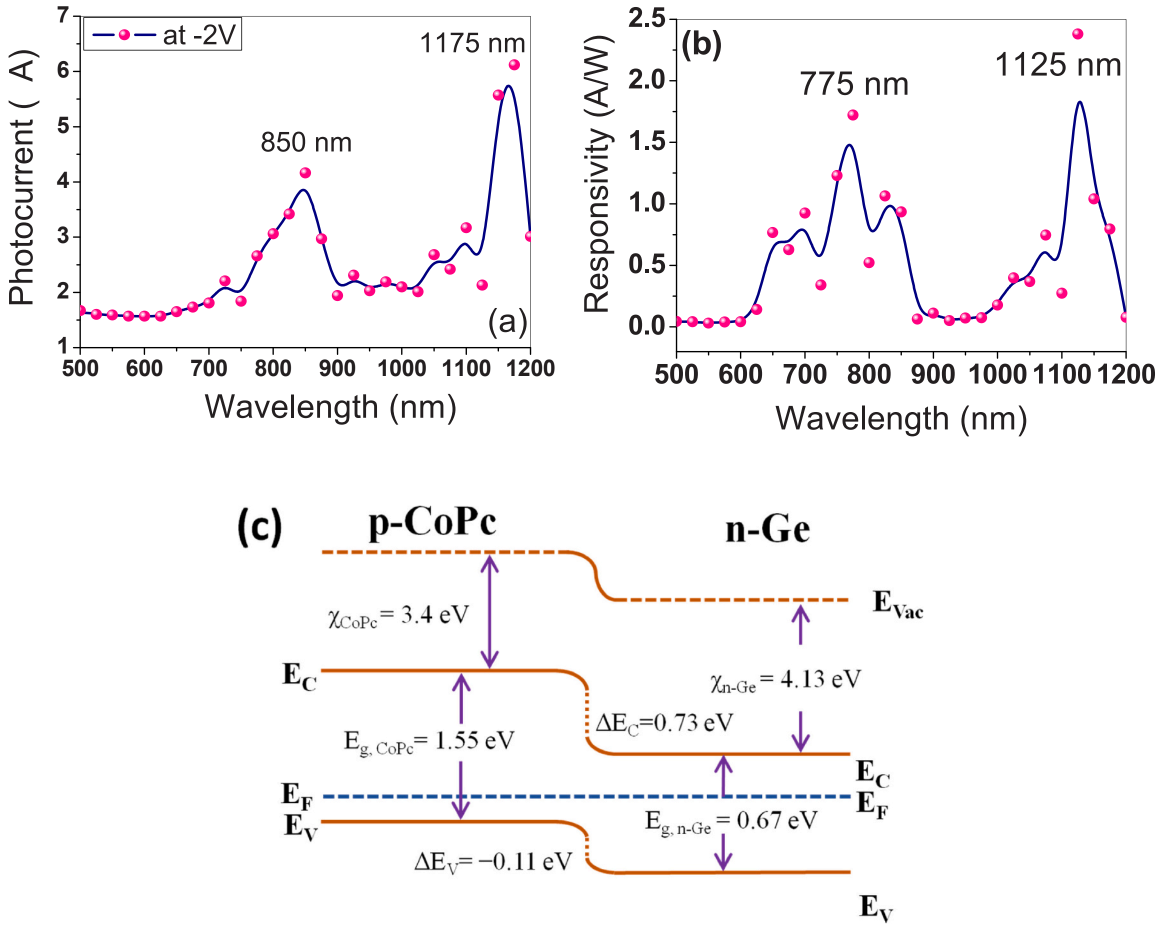

Fig. 2. Analysis of Photodiode parameters such as (a) Photocurrent (b) Responsivity at −2 V reverse bias voltage and (c) Energy band diagram of CoPc/Ge heterojunction.

maximum responsivity was observed at 775 nm (for a normalized power of 0.922 $\mu W/cm^2$) and 1125 nm (for a normalized power of 2.19 $\mu W/cm^2$) with 1.72 A/W and 2.38 A/W respectively. Other photodiode parameters like detectivity and EQE relied on responsivity hence the detectivity found similar variation as like resposnivity with wavelength. Fig. 3(a) describes the detectivity of CoPc/n-Ge heterostructure at varied illumination wavelengths. The magnitude of detectivity at 775 nm is found to be 1.09×10^{11} Jones and

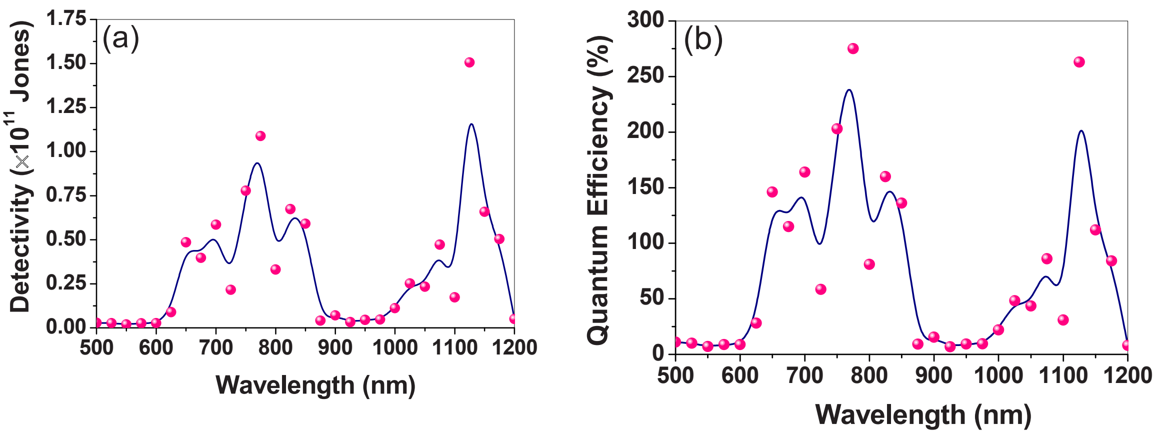

Fig. 3. Variation of (a) Detectivity and (b) Quantum efficiency at reverse bias voltage of −2 V under illumination wavelengths varied from 500 nm to 1200 nm in steps of 25 nm.

1.51 $\times 10^{11}$ Jones at 1125 nm illumination wavelength. Notably, the EQE of CoPc/n-Ge heterostructure depicted in Fig. 3(b) is found to be 275 % at illumination wavelength of 775 nm and 263% at 1125 nm. Larger EQE greater than 100 % observed in CoPc/Ge heterojunction might be associated due to well defined interface between the layers of the heterojunction. This may also be attributed to narrowing of the depletion region due to localization of photogenerated minority carriers at the semiconductor surface. Similar observations were also indicated in the earlier reports [31,32]. The photodiode parameters observed in the present work are very much comparable with the earlier reports which found that the CoPc layer on n-Ge significantly enhances the responsivity to illumination wavelengths even at significantly low optical power [22,33,34].

Further analysis of barrier properties of the CoPc/n-Ge heterojunction under illumination was investigated at dark, 775 nm, 850 nm, 1125 nm and 1175 nm wavelengths. These selected illumination wavelengths show either maximum photocurrent or else large photoresponsivity at larger reverse voltage of −2 V. The dark and photoresponsive I-V characteristics of the Au/CoPc/n-Ge Schottky contacts under selected illumination wavelengths are shown in Fig. 4(a). For the CoPc/n-Ge heterojunction photodiode, assuming thermionic emission theory, the barrier height and ideality factor values are determined from ln (I) vs. voltage plot (not shown here) as described by Yildiz et al. [35]. The barrier height (Φ_b) and ideality factor (n) values were found to be 0.73 eV and 1.33 at dark respectively. On the other hand, Φ_b and n magnitudes were found to be 0.68, 0.71, 0.71, 0.71 and 1.40, 1.27, 1.34, and 1.32 at the illumination wavelengths of 775, 850, 1125, 1175 nm, respectively. It has been observed that the barrier height values are higher under dark than compared to other illumination wavelengths. The barrier height is minimum at the wavelength of 775 nm for the PDs, where the highest photoresponse is attained. It is observed that the photocurrent is higher than the dark current which implies the spectral response for the illuminated wavelengths [19].

Fig. 4(b) depicts the junction resistance (R_J) versus voltage (V) graph of the Au/CoPc/n-Ge heterojunction PD for dark, 775 nm, 850 nm, 1125 nm and 1175 nm. The R_s and R_{sh} are the important parameters which influence the performance of the heterojunction PD. The series resistance of the heterostructure was determined at large forward voltages where the resistance reaches to a lower magnitude. The evaluated series resistance (R_s) values are found to be 2654 Ω, 1309 Ω, 1320 Ω, 1197 Ω and 1324 Ω for dark, 775 nm, 850 nm, 1125 nm and 1175 nm respectively. Whereas high resistances observed at zero bias voltage leads to finding shunt resistances (R_{sh}). The R_{sh} values of the contacts were found to be 1.37$\times 10^6$ Ω, 4.53$\times 10^5$ Ω, 2.70$\times 10^5$ Ω, 5.85$\times 10^5$ Ω, 8.06$\times 10^5$ Ω for dark, 775 nm, 850 nm, 1125 nm and 1175 nm, respectively. The influence of junction resistance (both R_S and R_{sh}) of the CoPc/n-Ge heterostructure at different illumination wavelengths was found to be negligible. However, the junction resistance of the heterostructure seems to be decreased by two orders of magnitude at illumination than compared to dark. This might be due to increased charge carrier generation at the junction under illumination causes to produce more photocurrent and hence reduction in junction resistance.

The barrier parameters like Φ_b, n are influenced by the series resistance (R_S) which is most significant at high forward current. Hence, the deduction of series resistance at high forward current seems to be very much essential. A very standard methodology to evaluate series resistance has been utilized as proposed by Cheung and Cheung [36]. According to Cheung's plot, at forward bias voltages, the plot of dV/dlnI vs. I and H(I) vs. I plot always provides the parameters like R_S from the slope and n and Φ_b from the intercept of the plots, respectively. Fig. 5 and 6 shows the dV/dlnI vs. I and H(I) vs. I plots of the CoPc/n-Ge heterostructure under dark and various selective illumination wavelengths. The R_S and n values are determined from the slope and intercept of the dV/dlnI vs. I graph (Fig. 5). The R_s and n values are found to be 2718 Ω, 1765 Ω, 2074 Ω, 1652 Ω and 1687 Ω and 3.09, 2.66, 1.94, 2.11 and 2.08 for dark, 775 nm, 850 nm, 1125 nm and 1175 nm respectively. As shown in Fig. 6, the H(I) vs. I plot revealed a straight line with the current axis intercept equal to nΦ_b. The R_s and Φ_b values are found to be 2679 Ω, 1734 Ω, 2009 Ω, 1596 Ω and 1615 Ω and 0.68, 0.64, 0.68, 0.68 and 0.69 eV for dark, 775 nm, 850 nm, 1125 nm and 1175 nm. These results indicate the consistency in the determination of R_S from Cheung's methods with R_J versus voltage method. Further, the determination of other barrier parameters using Cheung's methods might also be consistent with thermionic emission approach. The R_S measured at 775 nm was found smaller than dark by two orders of magnitude which might be due to carrier generation at the junction because of illumination. On the other hand, R_S is found to be quite sensitive to the illumination wavelengths due to the effect of photogenerated carriers [37].

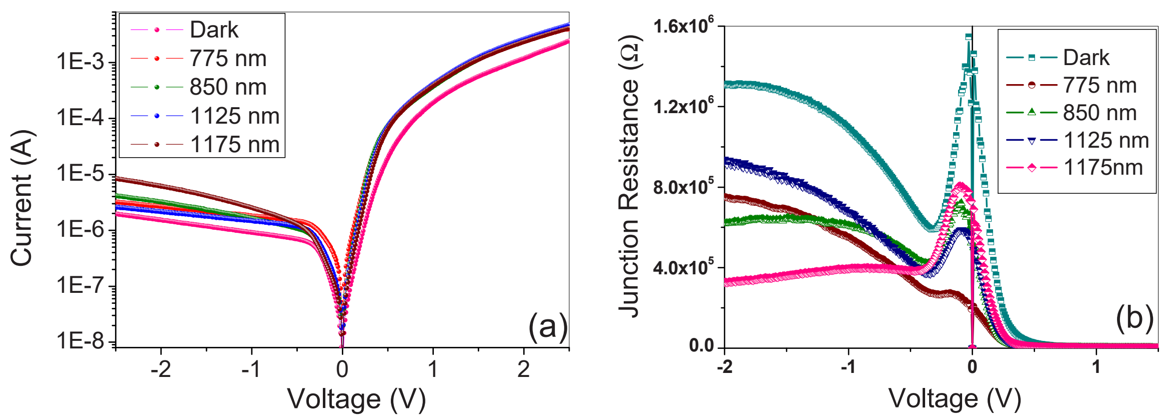

Fig. 4. (a) Electrical properties of CoPc/n-Ge heterojunction photodiode (b) Junction resistance versus voltage of the photodiode for the selective wavelengths (dark, 775 nm, 850 nm, 1125 nm and 1175 nm).

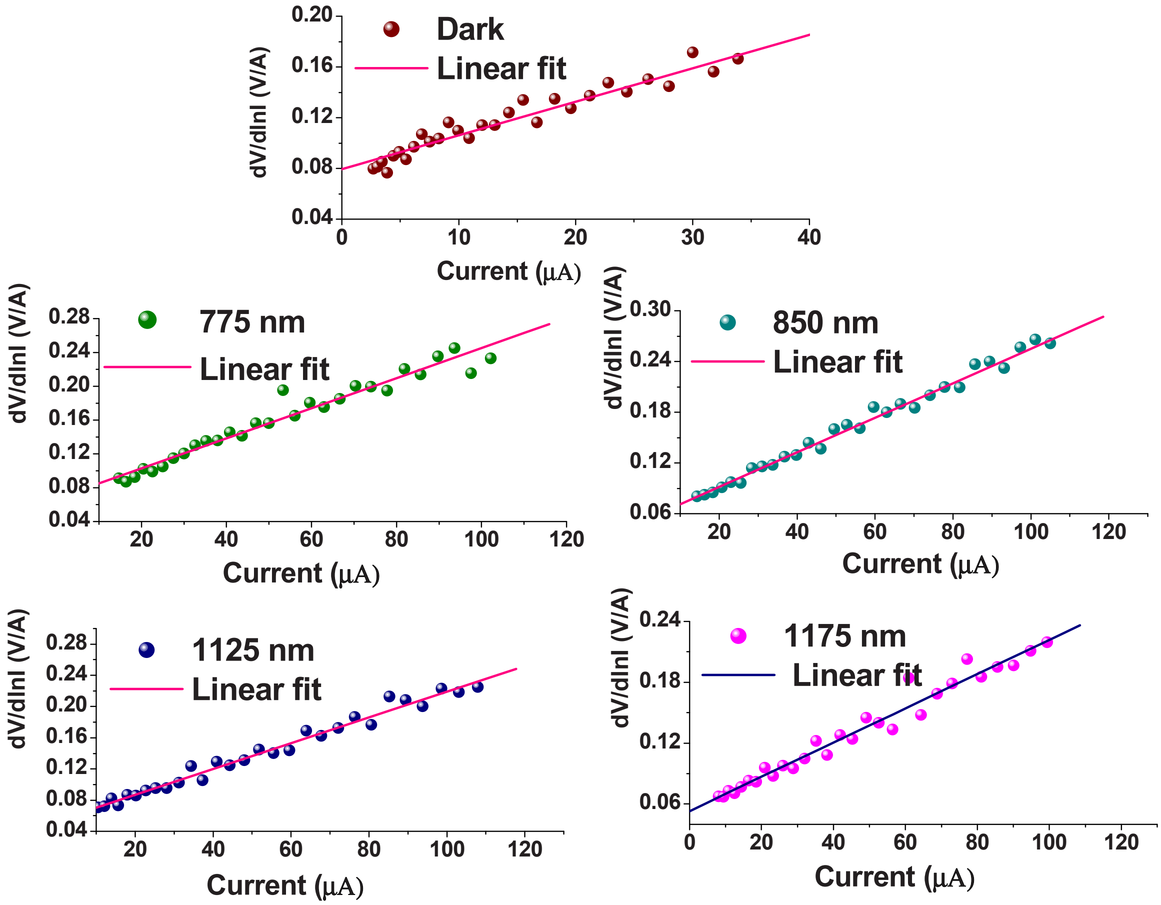

Fig. 5. Cheung's plot ((dV/dlnI) versus forward current) of CoPc/n-Ge heterojunction photodiode for the selective wavelengths.

An alternative method is also proposed by Norde [38] to evaluate barrier parameters like Φ_b and R_S in the forward bias regime. According to Norde, a plot of F(V) versus V (as shown in Fig. 7) defines a minimum point of voltage and current (V_{min} and I_{min}) for the corresponding F(V_{min}). The value of V_{min}, I_{min} and F(V_{min}) might be helpful to evaluate Φ_b and R_S. The barrier parameters are evaluated under dark and at various illumination wavelengths. The R_S and Φ_b values are found to be 74455 Ω, 8497 Ω, 35106 Ω, 43996 Ω and 49656 Ω, 0.74 eV, 0.68 eV, 0.71 eV, 0.71 eV and 0.71 eV for dark, 775 nm, 850 nm, 1125 nm and 1175 nm, respectively. Table 1 describes the barrier parameters evaluated using different methods for the CoPc/n-Ge heterostructure under dark and illumination conditions. The Φ_b evaluated from the Norde method shows good similarity with other approaches such as ln I vs. voltage approach and Cheung's method. However, the series resistance was found higher in the Norde approach than compared to other methods which might be associated with the region-dependent evaluation of parameters by these methods. In general, the Norde method hopefully works well for ideal rectifying junctions and in turn the barrier parameters are evaluated for the complete range of I-V data. Due to the non-ideal nature of the contacts, the series resistance observed in the Norde approach was found to be larger than compared to other methods [39,40].

A detailed investigation of the conduction behaviour of the heterostructure in the whole forward bias regime is very important as it define the device's performance. Fig. 8(a) represents the log (I) vs. log (V) plot to determine the current conduction mechanism in the forward bias region of the CoPc/n-Ge heterostructure under selected illumination wavelengths. It is evident from Fig. 8(a) that the log (I) vs. log (V) graph has three distinct linear regions (named as region- I, region-II and region-III) with different slope factors (m). The slope factor of the linear regions is significantly influenced by the deep traps presented at the junction causes to change in the mobility of the charge carriers during current conduction. Further, the carrier mobility across the junction is also influenced by bias. Hence, these three linear regions obey the $I \sim V^m$ relation where the current is proportional to the applied bias voltage with a power factor of m [41,42]. The slope factor (m) evaluated from the different regions was plotted against the illumination conditions as shown in Fig. 8 (b) (magnitudes of m are mentioned inside the graph at each data point). The region-I is defined from 0 V to 0.14 V where the slope factors varied from 1.36, 1.29, 1.4, 1.23 and 1.27 for dark, 775 nm, 850 nm, 1125 nm and 1175 nm respectively. The variation of m in region-I signifies the current conduction dominant in the mentioned voltage range for dark and illumination conditions might be associated with the ohmic conduction. Due to this conduction mechanism, the primary current transport in this region may be caused by thermionic emission (TE) current whereas the injection current is found to be negligible due to low bias. In region-II

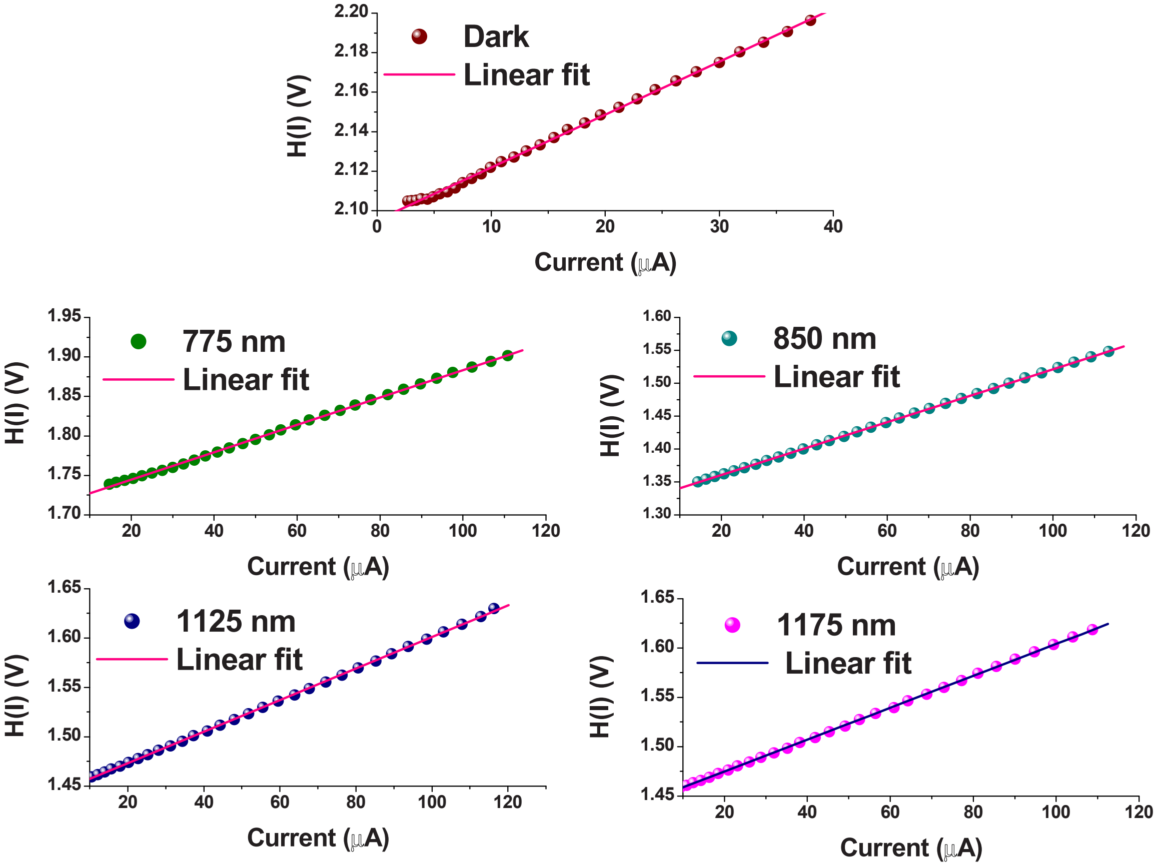

Fig. 6. Cheung's function (H(I)) versus forward current of CoPc/n-Ge heterojunction photodiode for the selective wavelengths.

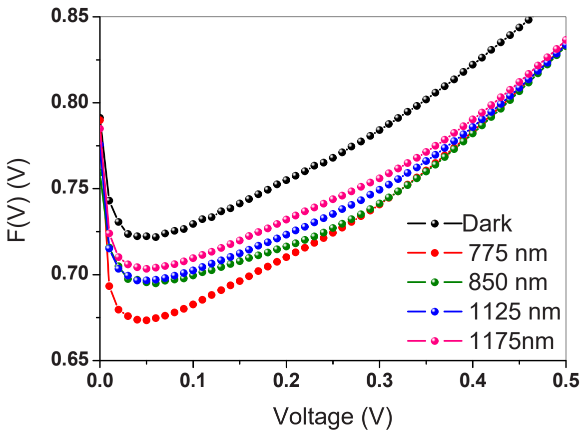

Fig. 7. Norde function (F(V)) versus voltage of CoPc/n-Ge heterojunction photodiode for the selective wavelengths.

(0.14<V<0.44), the slope factors for dark and illumination wavelengths (775 nm, 850 nm, 1125 nm and 1175 nm) are found to be 3.69, 3.27, 3.75, 3.49 and 3.7, respectively. At moderately high voltages, the slope factor varied from 3.69 to 3.7 signifies the current conduction can be caused by space charge limited current (SCLC) assisted by traps. For the moderate bias range, the increased density of carriers by bias and illumination might be injected through the traps and traps tends to be filled up which built up the space charges at the depletion region [42]. At a sufficiently high bias range in region-III (0.44<V<3), the slope factor varies from 2.89 to 2.58 indicating that the value of the slope factor decreased from region-II. This might be due to strong carrier injection under high field; the carriers tend to escape from traps and carrier-free traps causes to exhibit space charges at the junction [42]. Hence, the current

Table 1
Barrier properties of CoPc/n-Ge heterojunction photodiode using different methods for the selective illumination wavelengths.

Illumination wavelength	ln I-V		Junction Resistance Method		Cheung's 1 Method		Cheung's 2 Method		Norde Method	
	Φ_B (eV)	n	R_{sh} (Ω)	R_S (Ω)	Rs (Ω)	n	Rs (Ω)	Φ_B (eV)	Rs (Ω)	Φ_B (eV)
Dark	0.73	1.33	1.37×10^6	2654	2718	3.09	2679	0.68	74455	0.74
775	0.68	1.40	4.53×10^5	1309	1765	2.66	1734	0.64	8497	0.68
850	0.71	1.27	2.70×10^5	1320	2074	1.94	2009	0.68	35106	0.71
1125	0.71	1.34	5.85×10^5	1197	1652	2.11	1596	0.68	43996	0.71
1175	0.71	1.32	8.06×10^5	1324	1687	2.08	1615	0.69	49656	0.71

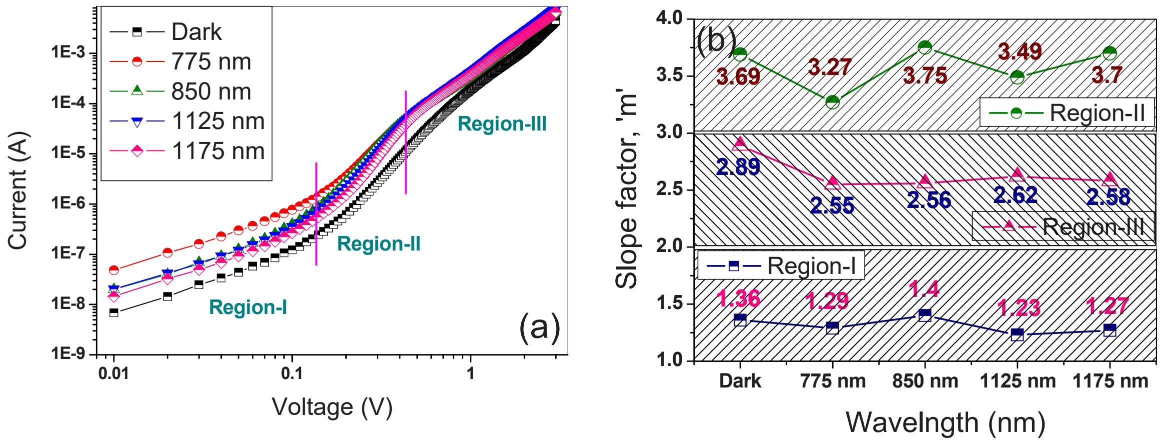

Fig. 8. Analysis of current conduction mechanisms in the forward bias (a) using log (I) versus log (V) plot (b) Variation of slope factor 'm' under different regimes for various selective wavelengths.

conduction observed in region-III might be due to SCLC.

Photodiode properties are generally significant in the reverse bias regime of the I-V properties of the heterostructure. The effectiveness of the SCLC mechanism is justified by the density of traps and their energy levels. Understanding of different transport processes associated with the reverse bias like Poole-Frenkel emission (PFE) and Schottky emission (SE) should also be essential to knowing the carrier transport across the junction. In the reverse bias regime, two field-lowering mechanisms such as PFE and SE were analysed [43]. Considering SE and PFE theory, using the relation between reverse current (I_R) and reverse voltage (V_R) defined in earlier reports [42], a plot has been drawn between ln(I_R) versus $\sqrt{V_R}$ as shown in Fig. 9. The theoretical slopes were evaluated using the relation mentioned [42] by considering the permittivity of the CoPc as 3.6 [26]. The theoretical field lowering coefficients such as β_{SE} (for SE) and β_{PF} (for PFE) are found to be $\sim2\times10^{-5}$ $eVm^{1/2}V^{-1/2}$ and $\sim4\times10^{-5}$ $eVm^{1/2}V^{-1/2}$ respectively. The figure reveals two linear regions for the heterostructures under dark and illumination with experimental slopes mentioned in Table 2. The region-I observed at lower bias of the CoPc/n-Ge heterojunction under illumination and dark shows the experimental slopes which are nearly matched with β_{PF}. On the other hand, the slopes measured in region-II for the heterojunction under illumination and dark nearly matched with β_{SE}. This suggests that PFE and SE transport processes are actively taking part in producing the reverse current in the heterostructures.

The presence of interface states and their distribution at the CoPc/Ge interface under dark and illumination also effectively modifies the current transport across the junction. According to Card and Rhoderick, the density of states (N_{ss}) and the bias-dependent ideality factor are related as mentioned [41,44]. The energy of states below the conduction band (E_C-E_{SS}) is represented with respect to N_{ss}. Fig. 10 represents the distribution of states with E_C-E_{SS} for the heterostructure under dark and illumination. The interface states are presented above the mid-gap of the n-Ge semiconductor and are found to decrease with increasing E_C-E_{SS}. The distribution of interface states was observed from E_C-0.4 eV to E_C-0.72 eV with the density of states varied from 2.28×10^{12} $eV^{-1}cm^{-2}$ to 2.35×10^{11} $eV^{-1}cm^{-2}$, respectively. It is also evidenced that the density of states was found to be decreased under illumination compared to dark. As the distribution of states exists from the mid-gap of the semiconductor, the states are presumably shallow (donor-type). Under illumination, a large density of carriers present at the junction can migrate across the junction under large fields. The shallow surface states can be neutralized by trapping the carriers across the junction during their migration. This effectively reduces the density of states at the interface under illumination. These observed results are very much comparable with the earlier reports [41,42].

4. Conclusion

The forward and reverse bias current-voltage measurements of fabricated the Au/CoPc/n-Ge photodiode have been investigated in

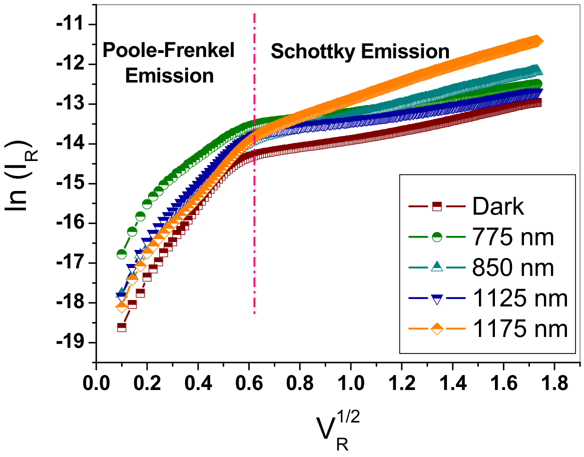

Fig. 9. Exploring the various reverse current conduction mechanisms of CoPc/n-Ge heterojunction photodiode for the selective wavelengths.

Table 2
Evaluation of empirical magnitudes of reverse conduction mechanisms of CoPc/n-Ge photodiode under different selective wavelengths.

Illumination wavelength	Theoretical field lowering constants		Experimental Slope Magnitudes	
	β_{SC} ($eVm^{1/2}V^{-1/2}$)	β_{PF} ($eVm^{1/2}V^{-1/2}$)	Region-I	Region-II
Dark			6.02×10^{-5}	8.34×10^{-6}
775 nm			3.23×10^{-5}	6.30×10^{-6}
850 nm	2.00×10^{-5}	4.00×10^{-5}	4.63×10^{-5}	1.07×10^{-5}
1125 nm			4.69×10^{-5}	6.51×10^{-6}
1175 nm			4.92×10^{-5}	1.46×10^{-5}

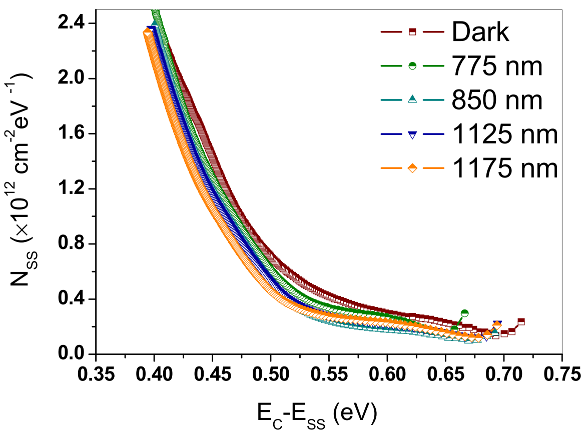

Fig. 10. Interface state density versus surface state energy below conduction band plot of heterojunction photodiode for the selective wavelengths.

the dark and under various illumination wavelengths. The heterostructure shows high rectification characteristics under various illumination wavelengths. CoPc/n-Ge heterojunction photodiode (at −2 V) shows better responsivity, detectivity and quantum efficiency at 775 nm and 1125 nm illumination wavelength than compared to other wavelengths. Barrier parameters of the heterostructure at selective wavelengths (775 nm, 850 nm, 1125 nm and 1175 nm) reveal noticeable variation. Further, the forward conduction processes of this heterojunction photodiode identify with three regions. At moderately high voltages, the charge transport is mainly caused by the TCLC process. The reverse conduction processes were also evaluated for the photodiode which represents two regions, region-I is identified at low voltages with PFE and the SE process is dominant at high reverse voltages (region-II). The interface state density of the photodiode for the selective illumination wavelengths is found to be decreased with surface state energy below the conduction band. Further, the decreased interface state density is observed with illumination wavelength at any surface state energy which might be associated with the trapping of charge carriers by the shallow states presented at the CoPc/n-Ge interface.

CRediT authorship contribution statement

M. Pavani: Writing – review & editing, Validation, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **A. Ashok Kumar:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Methodology, Formal analysis, Data curation, Conceptualization. **V. Rajagopal Reddy:** Writing – review & editing, Validation, Resources, Methodology, Formal analysis, Data curation. **S. Kalleemulla:** Writing – review & editing, Validation, Resources, Formal analysis. **V. Janardhanam:** Writing – review & editing, Validation, Resources, Methodology, Formal analysis, Data curation. **Chel-Jong Choi:** Writing – review & editing, Validation, Resources, Methodology, Formal analysis, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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